

25 *User Stories* Full System Coverage

Covering all 10 tables across ML prediction, data analysis, Power BI visualisation, and MongoDB / MySQL storage — refined with proper acceptance criteria.

25

7

10

USER STORIES

ROLES

TABLES

STORY TYPE →

ML / Prediction

Data Analysis

Power BI

MySQL

MongoDB



City Planners & Water Board Officers

Strategic planning, demand forecasting, infrastructure investment decisions

US-01 → US-04

US-01

Predict City-Wide Daily Water Demand for Next 12 Weeks

 Water Board Planning Officer

As a **Water Board Planning Officer**, I want to **predict daily water demand for the next 12 weeks per city**, so that I can pre-allocate reservoir supply and prevent shortages before they occur.

ACCEPTANCE CRITERIA

- GBT or XGBoost model trained on `demand_lag_1w`, `demand_roll_12w_avg`, `seasonal_factor`, `rainfall_mm`, `rain_3m_sum`, `is_drought_year`
- Time-based train/test split: 2019–2023 train, 2024 holdout; MAPE < 10% on holdout
- 12-week predictions saved per `city_id` and `date` in MySQL `predictions` table
- SHAP values computed to explain which features drive demand spikes

KEY COLUMNS

<code>estimated_demand_mld</code>	<code>demand_lag_1w</code>	<code>demand_lag_2w</code>	<code>demand_roll_12w_avg</code>
<code>seasonal_factor</code>	<code>rainfall_mm</code>	<code>rain_3m_sum</code>	<code>is_drought_year</code>
<code>is_monsoon_month</code>	<code>city_id</code>		

 ML Prediction

MySQL

HIGH

US-02

Power BI: City-Level Supply vs Demand Gap Dashboard

 Municipal Commissioner

*As a **Municipal Commissioner**, I want a **live Power BI dashboard** showing **supply vs demand gaps across all 15 cities**, so that I can *prioritise budget allocation and inter-city water sharing decisions in real time.**

ACCEPTANCE CRITERIA

- Bar chart of `daily_supply_mld` vs `daily_demand_mld` with `supply_deficit_mld` highlighted in red per city
- KPI cards showing `avg_hours_supply_per_day` and `nwr_pct` per city with traffic-light conditional formatting
- Heatmap of `nwr_pct` by city and `zone` (North/South/East/West/Central)
- Slicers on `state`, `climate_zone`, and `year` for drill-down

KEY COLUMNS

supply_deficit_mld

daily_supply_mld

daily_demand_mld

nrv_pct



Power BI



Analysis

HIGH

US-03

Store & Query Water Source Utilisation History in MongoDB



Source Management Engineer

*As a **Source Management Engineer**, I want to **store all water source records in MongoDB and query utilisation trends**, so that I can track capacity degradation over years and flag sources operating below safe capacity.*

ACCEPTANCE CRITERIA

- MongoDB collection `water_sources` stores all fields including `installed_capacity_mld`, `current_capacity_mld`, `source_type`, `is_active`
- Query: find all sources where `current_capacity_mld` < 70% of `installed_capacity_mld` — returns source name, city, and deficit MLD
- Compound index on `city_id` + `source_type` for queries returning results in < 100ms
- Filter support: `is_active`, `treatment_plant_present`, `year_commissioned` range queries

KEY COLUMNS

`source_type`

`installed_capacity_mld`

`current_capacity_mld`

`is_active`

`distance_from_city_km`

`year_commissioned`

`last_upgraded_year`

`treatment_plant_present`

MongoDB



Analysis

MEDIUM

US-04

Predict 2030 Population-Driven Water Demand by City



Urban Infrastructure Planner

*As an **Urban Infrastructure Planner**, I want to **project water demand per city through 2030 based on population growth**, so that I can plan reservoir expansions*

and pipeline capacity upgrades before the deficit becomes critical.

ACCEPTANCE CRITERIA

- Use `population`, `annual_growth_rate_pct`, and `actual_lpcd` to compute `est_daily_water_demand_mld` for 2025–2030
- Flag cities where projected demand will exceed sum of `installed_capacity_mld` of all active sources
- `projected=True` rows clearly separated from historical records in MySQL output
- Power BI line chart showing projected demand trajectory vs current supply capacity per city

KEY COLUMNS

`population`

`annual_growth_rate_pct`

`est_daily_water_demand_mld`

`projected`

`actual_lpcd`

`installed_capacity_mld`

`current_capacity_mld`



ML Prediction

MySQL



Power BI

HIGH

B Field Engineers & Operations Teams

Pipeline maintenance, anomaly detection, disruption analysis

US-05 → US-08

US-05

Predict Pipeline Burst Risk by Ward Using Infrastructure Age



Pipeline Maintenance Engineer

As a Pipeline Maintenance Engineer, I want to predict which wards are at high risk of pipeline burst in the next 3 months, so that I can schedule preventive maintenance before the monsoon season causes failures.

ACCEPTANCE CRITERIA

- Binary classifier using `pipe_age_years`, `pipeline_condition_encoded`, `estimated_leakage_pct`, `pipeline_material`, `last_major_repair_year` as features
- Training labels derived from `fact_disruptions` where `cause = 'Pipeline Burst'` joined to `ward_id`
- Hard rule: any ward with `pipe_age_years` > 40 AND `pipeline_condition` = Poor automatically flagged Critical regardless of model output
- Risk scores per `ward_id` stored in MongoDB `risk_scores` collection with timestamp

KEY COLUMNS

`pipe_age_years` `pipeline_condition_encoded` `estimated_leakage_pct`
`pipeline_material` `last_major_repair_year` `cause` `mat_AC` `mat_CI`



MongoDB

HIGH

US-06

Detect Supply Anomalies Where Efficiency Drops Below 50%

Operations Monitoring Analyst

As an Operations Monitoring Analyst, I want an anomaly detection model that flags wards with supply efficiency below 50%, so that field crews can be dispatched immediately before residents face prolonged outages.

ACCEPTANCE CRITERIA

- Use `is_anomaly` (pre-built in `fact_supply`) as target; features: `supply_efficiency_pct`, `hours_of_supply`, `nwr_vs_benchmark`, `supply_roll_4w_avg`
- Minority class oversampled (anomalies are rare) using SMOTE or random oversampling before training
- False positive rate < 5%; precision on anomaly class > 85%
- Detected anomalies pushed to MongoDB `alerts` collection with `ward_id`, `date`, `city_id`, and predicted probability

KEY COLUMNS

`is_anomaly` `supply_efficiency_pct` `hours_of_supply` `nwr_vs_benchmark`

supply_roll_4w_avg

has_scada_monitoring

supply_lag_1w



MongoDB

HIGH

US-07

Power BI: Infrastructure Health Map Across All Wards



Infrastructure Asset Manager

*As an **Infrastructure Asset Manager**, I want a **ward-level Power BI map** showing **pipeline condition, leakage %, and pipe age**, so that I can objectively prioritise the annual infrastructure renewal budget across 1,809 wards.*

ACCEPTANCE CRITERIA

- Choropleth map coloured by `pipeline_condition` (Good=green, Fair=yellow, Poor=red)
- Bubble overlay sized by `estimated_leakage_pct` — bigger bubble = worse leakage
- Ranked table showing wards where `pipe_age_years` > 40, sorted descending by `estimated_leakage_pct`
- Slicers for `pipeline_material`, `last_major_repair_year` range, and `city_id`

KEY COLUMNS

`estimated_leakage_pct`

`pipe_age_years`

`pipeline_condition`

`pipeline_material`

`last_major_repair_year`

`num_pumping_stations`

`storage_tank_capacity_kl`



Power BI



Analysis

HIGH

US-08

Analyse Disruption Patterns and Resolution Time by Cause



Operations Control Room Manager

*As an **Operations Control Room Manager**, I want to **analyse disruption events grouped by cause and severity** to find which types take longest to resolve, so that I can staff response teams appropriately and reduce mean resolution time.*

ACCEPTANCE CRITERIA

- Group `fact_disruptions` by `cause` : compute avg `duration_hours` , total `population_affected` , sum `complaint_count` , and `resolved` rate
- Severity distribution (Critical/High/Medium/Low) broken down per city and per `cause`
- Most common `resolution_action` mapped to each `cause` — identifies best-practice response procedures
- Monthly aggregation stored in MySQL `disruption_summary` for ops review meetings

KEY COLUMNS

`cause` `duration_hours` `severity` `population_affected` `complaint_count`
`resolved` `resolution_action` `num_wards_affected`



Analysis

MySQL

MEDIUM



Data Scientists & ML Engineers

US-09 → US-12

Model building, clustering, forecasting, advanced ML pipelines

US-09

Classify Ward Deficit Severity (None / Low / Medium / High)



ML Engineer

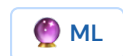
As an ML Engineer, I want to build a multi-class classifier predicting `deficit_severity` per ward-week, so that water boards can proactively trigger tanker deployments before a deficit becomes Critical.

ACCEPTANCE CRITERIA

- Target: `deficit_severity` (0=None, 1=Low, 2=Medium, 3=High) from `fact_supply`
- Features: `demand_roll_4w_avg`, `supply_efficiency_pct`, `rainfall_mm`, `is_drought_year`, `ward_type_encoded`, `leakage_pct`
- Class imbalance addressed: most records are class 0 — use class weights or SMOTE
- Model outputs (class + probability) stored per record in MongoDB `ml_outputs` collection

KEY COLUMNS

`deficit_severity` `supply_efficiency_pct` `demand_roll_4w_avg` `rainfall_mm`
`is_drought_year` `ward_type_encoded` `rain_3m_sum`



MongoDB

HIGH

US-10

Cluster Wards by Water Access Profile (K-Means)

Data Scientist

As a Data Scientist, I want to cluster wards into water access profiles using K-Means, so that government welfare schemes can be targeted at the most deprived clusters rather than applied uniformly.

ACCEPTANCE CRITERIA

- Features: `piped_connection_coverage_pct`, `metered_connections_pct`, `has_slum_pocket`, `ward_type_encoded`, avg `supply_efficiency_pct`, avg `hours_of_supply`
- Optimal K validated using silhouette score across K=3 to K=6; best K selected automatically
- Cluster labels joined back to `ward_id` and stored in MySQL `ward_clusters` table
- Power BI pie chart showing cluster distribution; each cluster named (e.g. "Well-Served", "Underserved", "Slum-Heavy")

KEY COLUMNS

`piped_connection_coverage_pct` `metered_connections_pct` `has_slum_pocket`

ward_type_encoded

avg_hours_supply_per_day

supply_efficiency_pct



MySQL



MEDIUM

US-11

Time-Series Forecasting of LPCD Trends by Ward Type



Demand Forecasting Analyst

As a **Demand Forecasting Analyst**, I want to **forecast LPCD trends per ward type using weekly historical data**, so that I can identify whether Slum/Informal wards are getting better or worse access over time and escalate if the equity gap widens.

ACCEPTANCE CRITERIA

- One model per **ward_type** using **lpcd_used** as target with **date** as time index (weekly frequency)
- External regressors: **is_festival_month**, **covid_adjustment**, **season**, **rainfall_mm**
- Slum ward LPCD compared against Residential at each forecast step — alert if gap exceeds 80 LPCD
- 12-week forecast output stored per **ward_type** in MySQL **lpcd_forecast** table

KEY COLUMNS

lpcd_used

ward_type

date

is_festival_month

covid_adjustment

season

demand_lag_1w



MySQL

MEDIUM

US-12

Predict Whether Tanker Supplement Will Be Needed Next Week



Supply Logistics ML Engineer

As a **Supply Logistics ML Engineer**, I want to **predict whether a ward will need tanker supplementation next week**, so that tanker fleets can be pre-positioned before the deficit occurs rather than reactively deployed.

ACCEPTANCE CRITERIA

- Binary target: `is_tanker_supplement` ; features: `supply_deficit_mld` , `deficit_roll_4w` , `hours_of_supply` , `has_slum_pocket` , `is_drought_year`
- Precision > 80% on tanker=1 class — cost of missing a tanker deployment (false negative) is high
- Model retrained weekly; next-week predictions pushed to MongoDB `tanker_schedule` collection daily
- Output includes ward, city, predicted probability, and recommended tanker count based on deficit volume

KEY COLUMNS

`is_tanker_supplement` `supply_deficit_mld` `deficit_roll_4w` `hours_of_supply`
`has_slum_pocket` `is_drought_year` `population_2019`



MongoDB

HIGH

D Policy Makers & Government Analysts

US-13 → US-16

NRW tracking, tariff equity, COVID impact, compliance reporting

US-13

Analyse COVID-19 Impact on Water Demand Across Ward Types



Government Policy Analyst

*As a **Government Policy Analyst**, I want to quantify how COVID-19 lockdowns in 2020 impacted water demand by ward type, so that I can include water utility resilience requirements in future national disaster policy frameworks.*

ACCEPTANCE CRITERIA

- Compare `estimated_demand_mld` where `covid_adjustment=1` (Apr–Jun 2020) vs same months in 2019 and 2021 — compute % change
- Segment by `ward_type`: Commercial and Industrial wards expected to show largest demand drop
- Secondary metrics: change in `hours_of_supply` and `nrv_pct` during lockdown period vs baseline
- Summary table stored in MySQL `policy_reports` with city, ward_type, period, and % impact

KEY COLUMNS

`covid_adjustment` `estimated_demand_mld` `ward_type` `year` `month`
`hours_of_supply` `nrv_pct`



Analysis

MySQL

MEDIUM

US-14

Power BI: Tariff Equity Analysis — Is Water Affordable for the Poor?



Water Tariff Reform Officer

As a Water Tariff Reform Officer, I want a Power BI report comparing tariff structures across all 15 cities, so that I can identify cities where poor households face a disproportionately high cost burden for basic water access.

ACCEPTANCE CRITERIA

- Stacked bar: `rate_per_kl_inr` per `slab_number` per city — shows progressive vs regressive tariff structures
- Highlight Delhi's zero-rate Slab 1 (`rate_per_kl_inr = 0`) as a subsidy benchmark for other cities
- Scatter: `actual_lpcd` (x) vs Slab 1 `rate_per_kl_inr` (y) — cities in top-left quadrant (low LPCD + high rate) are equity red flags
- Fixed cost: `fixed_monthly_charge_inr` as % of city minimum wage — highlights affordability burden

KEY COLUMNS

`rate_per_kl_inr` `slab_number` `consumption_lower_kl` `fixed_monthly_charge_inr`
`actual_lpcd` `population_density_per_km2`



Power BI



Analysis

HIGH

US-15

Track NRW Reduction Progress and Flag Chronic Offender Wards



NRW Programme Director

As an NRW Programme Director, I want to track NRW % trends per city from 2019–2024 and flag wards that chronically exceed the city benchmark, so that I can direct leak reduction programmes at the highest-impact locations.

ACCEPTANCE CRITERIA

- Power BI line chart: `nwr_pct` trend over time per city, with city benchmark line overlaid
- Flag wards where `nwr_vs_benchmark` > 10 percentage points for 3 or more consecutive months
- Correlation analysis: `estimated_leakage_pct` × `pipe_age_years` × `metered_connections_pct` vs `nwr_pct`
- NRW analysis table stored in MySQL with ward and city granularity for monthly board reporting

KEY COLUMNS

`nwr_pct`

`nwr_vs_benchmark`

`city_nwr_benchmark`

`estimated_leakage_pct`

`metered_connections_pct`

`pipe_age_years`



Analysis



Power BI

MySQL

HIGH

US-16

Store All Disruption Audit Logs in MongoDB for CAG Compliance



Water Board Compliance Officer

As a Compliance Officer, I want all disruption records and resolution actions stored in MongoDB with an immutable audit trail, so that CAG audits and RTI queries can be answered instantly with a timestamped document history.

ACCEPTANCE CRITERIA

- MongoDB collection `disruption_audit`: every document includes `cause`, `severity`, `resolved`, `resolution_action`, `complaint_count`, `population_affected`
- Append-only collection: no update or delete operations permitted — every change creates a new document with version number
- Query: `resolved=False` + `city_id` + `year` returns all open unresolved disruptions for RTI compliance
- Index on `start_date` and `city_id` for sub-100ms query response on 5-year dataset

KEY COLUMNS

`resolved` `resolution_action` `cause` `severity` `complaint_count`
`estimated_supply_loss_mld` `population_affected` `start_date`

MongoDB



Analysis

MEDIUM

E Citizens, NGOs & Field Staff

US-17 → US-19

Equity analysis, complaint hotspots, public-facing insights

US-17

Power BI: Slum Ward Water Access Equity Report



NGO Water Rights Researcher

As an NGO Water Rights Researcher, I want a Power BI report comparing water access in Slum wards vs Residential wards across all cities, so that I can present concrete evidence of inequitable distribution when advocating for policy change.

ACCEPTANCE CRITERIA

- Side-by-side bar: `piped_connection_coverage_pct` and `metered_connections_pct` for each `ward_type` per city
- Box plot showing `lpcd_used` distribution for `wt_Slum/Informal=1` vs `wt_Residential=1` — visual proof of access gap
- Scatter: `has_slum_pocket` (x) vs `supply_efficiency_pct` (y) per city — shows if slum presence correlates with worse supply
- Regional breakdown by `zone` to identify which geographic regions have the worst slum equity gap

KEY COLUMNS

`wt_Slum/Informal` `piped_connection_coverage_pct` `metered_connections_pct`
`has_slum_pocket` `lpcd_used` `supply_efficiency_pct` `zone`



HIGH

US-18

Analyse Complaint Hotspots During Disruption Events

Citizens' Grievance Cell Analyst

*As a **Citizens' Grievance Analyst**, I want to **map disruption complaint hotspots by ward and identify seasonal complaint peaks**, so that the helpline can be adequately staffed during high-complaint periods and resources targeted to the right wards.*

ACCEPTANCE CRITERIA

- Rank wards by total `complaint_count` from all disruption events over 2019–2024
- Pearson correlation: `complaint_count` vs `duration_hours` and vs `population_affected` — quantify which drives complaints more
- Monthly heatmap: which `month` and `season` generate peak complaints — to inform staffing schedule
- Results stored in MySQL `complaint_analysis` table for monthly review meetings

KEY COLUMNS

`complaint_count` `duration_hours` `population_affected` `ward_id` `month` `year`
`cause` `severity`



Analysis

MySQL

MEDIUM

US-19

Predict Disruption Complaint Volume for Call Centre Staffing



Customer Relations Data Analyst

As a **Customer Relations Data Analyst**, I want to **predict how many complaints a disruption event will generate based on its characteristics**, so that call centre managers can pre-allocate staff and avoid overwhelming the grievance helpline.

ACCEPTANCE CRITERIA

- Regression target: `complaint_count` (log-transformed to handle skew); features: `duration_hours`, `population_affected`, `num_wards_affected`, `estimated_supply_loss_mld`, `cause` one-hot encoded
- $R^2 > 0.70$ on test set; RMSE reported in original complaint count units (reverse log-transform)
- Model prediction attached to each new disruption record in MongoDB at time of logging
- Output: predicted complaint count + 90% confidence interval for staffing buffer calculation

KEY COLUMNS

`complaint_count``duration_hours``population_affected``num_wards_affected``estimated_supply_loss_mld``cause`

ML

MongoDB

LOW



Climate & Hydrology Analysts

US-20 → US-22

Rainfall correlation, drought risk, flood impact analysis

US-20

Power BI: Monsoon Intensity vs Supply Efficiency Correlation



Hydrology Analyst

As a **Hydrology Analyst**, I want a **Power BI dashboard** showing how **monsoon rainfall intensity correlates with supply efficiency per city**, so that I can validate the rainfall-supply model assumptions and identify cities that are most vulnerable to monsoon variability.

ACCEPTANCE CRITERIA

- Dual-axis chart: monthly `rainfall_mm` bar vs avg `supply_efficiency_pct` line per city-year from 2015–2024
- Scatter: `departure_from_normal_pct` (x) vs `supply_deficit_mld` (y) — Pearson r shown on chart
- Background shading for `is_drought_year` (red) and `is_flood_year` (blue) periods on time axis
- Slicer by `climate_zone` to compare Arid (Jaipur) vs Tropical (Mumbai) city responses

KEY COLUMNS

`rainfall_mm``departure_from_normal_pct``is_drought_year``is_flood_year``rain_3m_sum``supply_efficiency_pct``climate_zone`

Power BI



Analysis

MEDIUM

US-21

Predict Drought Risk for Next Season Using Lagged Rainfall



Climate Risk Modeller

As a **Climate Risk Modeller**, I want to **predict whether a city will face drought-like supply conditions next quarter using lagged rainfall features**, so that water boards can activate rationing protocols 3 months in advance rather than reactively.

ACCEPTANCE CRITERIA

- Binary target: `is_drought_year` flag; features: `rain_lag_1m`, `rain_3m_sum`, `rain_6m_sum`, `rainy_days`, `departure_from_normal_pct`, `annual_rainfall_mm`
- Separate model trained per `climate_zone` (Arid, Semi-Arid, Tropical, Subtropical, Tropical Savanna)
- Recall > 80% on drought class — missing a drought year is more costly than a false alarm
- Quarterly predictions stored in MySQL `drought_risk_forecast` per city

KEY COLUMNS

`is_drought_year` `rain_lag_1m` `rain_3m_sum` `rain_6m_sum` `rainy_days`
`departure_from_normal_pct` `climate_zone`



MySQL

HIGH

US-22

Analyse Flood Year Impact on Disruption Frequency & Severity

Disaster Risk Analyst

As a Disaster Risk Analyst, I want to compare disruption frequency and severity in flood years vs normal years, so that I can build an evidence-based case for investing in flood-resilient water infrastructure.

ACCEPTANCE CRITERIA

- Join `dim_rainfall` (`is_flood_year=1`) with `fact_disruptions` on `city_id` + `year`
- Compare avg `duration_hours`, `num_wards_affected`, `estimated_supply_loss_mld` in flood years (2017, 2020, 2022) vs non-flood years
- Breakdown by `cause`: does "Monsoon Flooding" cause disproportionately more disruptions in flood years?
- Statistical t-test between flood vs non-flood disruption metrics — report p-value for significance

KEY COLUMNS

`is_flood_year` `duration_hours` `num_wards_affected` `estimated_supply_loss_mld`

cause year rainy_days

 Analysis

MongoDB

MEDIUM



Database Administrators & IT Teams

US-23 → US-25

Star schema design, MongoDB archiving, Power BI executive scorecard

US-23

Design MySQL Star Schema for Power BI Reporting Layer

 Database Architect

As a **Database Architect**, I want to **design and populate a MySQL star schema from the dim_ and fact_ CSV files**, so that *Power BI can connect via DirectQuery and analysts get sub-2-second query response on 568k rows.*

ACCEPTANCE CRITERIA

- Fact tables: `fact_demand`, `fact_supply`, `fact_disruptions` — all with FK constraints to dim tables
- Dim tables: `dim_cities`, `dim_wards`, `dim_sources`, `dim_infrastructure`, `dim_rainfall`, `dim_population`, `dim_tariff`
- Composite indexes on (`city_id`, `date`), (`ward_id`, `date`), and `source_id` for join performance
- DirectQuery benchmark: dashboard with 5 visuals must load in < 2 seconds on 568k rows

KEY COLUMNS

`city_id` `ward_id` `source_id` `demand_id` `supply_id` `rain_id` `tariff_id`
`infra_id` `date`

MySQL

 Power BI

HIGH

US-24

Archive Raw Fact Tables and ML Outputs in MongoDB Collections



Data Platform Engineer

As a **Data Platform Engineer**, I want to **store raw fact_demand and fact_supply records in MongoDB with ML outputs embedded**, so that the schema can evolve as new ML features are added without breaking the MySQL reporting layer.

ACCEPTANCE CRITERIA

- MongoDB collection **raw_demand**: all 568k records with **ward_type**, **city_name** embedded (denormalized) for fast reads
- MongoDB collection **raw_supply**: includes ML output fields **has_deficit**, **deficit_severity_label**, **is_anomaly**, **nwr_vs_benchmark**
- TTL index on **date**: records older than 7 years auto-archived to cold storage
- MySQL holds only aggregated city/ward summaries — raw records stay in MongoDB to separate OLTP from OLAP

KEY COLUMNS

has_deficit

deficit_severity_label

is_anomaly

nwr_vs_benchmark

supply_roll_4w_avg

demand_roll_4w_avg

date

MongoDB

MySQL

MEDIUM

US-25

Power BI Executive Scorecard: End-to-End Water Health per City



State Water Minister / Executive

As a **State Water Minister**, I want a **single-page executive Power BI scorecard per city showing overall water health**, so that I can make quarterly budget decisions and identify cities that need emergency intervention — in under 60 seconds.

ACCEPTANCE CRITERIA

- KPI cards per city: **actual_lpcd**, **nwr_pct**, **supply_deficit_mld**, **avg_hours_supply_per_day**, **metering_pct** with traffic-light RAG status

- Trend lines 2019–2024: `estimated_demand_mld` vs `actual_supply_mld` — shows whether gap is narrowing or widening
- Disruption summary box: total events, avg `duration_hours`, top `cause`, and unresolved count per city
- City rank table with composite water health score (weighted: LPCD 30%, deficit 30%, NRW 20%, hours 20%)

KEY COLUMNS

`actual_lpcd`

`nrrw_pct`

`supply_deficit_mld`

`avg_hours_supply_per_day`

`metering_pct`

`estimated_demand_mld`

`actual_supply_mld`

`piped_connection_coverage_pct`

 Power BI

 Analysis

 MySQL

HIGH

How to Implement Each Group of 5

Each section below maps to a group of user stories — follow these steps in order for each group.

1

ML Prediction Stories

US-01, 04, 05, 06, 09, 11, 12, 21

- 1 Load `ml_sample.csv` (56k rows) — your starting point for all ML models
- 2 Pick the target column for your model (e.g. `estimated_demand_mld` for US-01)
- 3 Split time-based: 2019–2023 = train, 2024 = test (never shuffle time-series data)
- 4 Train using PySpark MLlib GBT or sklearn — use the PySpark pipeline script already built
- 5 Evaluate: RMSE + R^2 for regression, AUC + F1 for classification. Save model to disk

2

Data Analysis Stories

US-08, 13, 15, 18, 22

- 1 Load the relevant table: `fact_supply.csv`, `fact_disruptions.csv`, `dim_rainfall.csv`
- 2 Use PySpark SQL `.groupBy()` + `.agg()` for aggregations — already shown in EDA steps of the pipeline
- 3 For correlation: use `df.stat.corr('col1', 'col2')` in PySpark or `df.corr()` in Pandas
- 4 Save output as CSV using `.toPandas().to_csv()` for MySQL import later
- 5 Import into MySQL using MySQL Workbench Table Import Wizard or Python `sqlalchemy`

3

Power BI Stories

US-02, 07, 14, 17, 20, 25

- 1 Open Power BI Desktop → Get Data → Text/CSV → load `dim_cities.csv`, `fact_supply.csv`, `dim_wards.csv`
- 2 In Model view: create relationships — `dim_cities[city_id]` → `fact_supply[city_id]` (one-to-many)
- 3 Build visuals: Bar chart = Demand vs Supply, KPI card = NRW%, Line chart = trend over time
- 4 Add slicers: `state`, `year`, `climate_zone`, `ward_type` for drill-down filtering
- 5 Publish to Power BI Service (free account) for shareable link or export as PDF for submission

4

MySQL Storage Stories

US-01, 04, 08, 13, 15, 23

- 1 Install MySQL Community Server (free) + MySQL Workbench (GUI)
- 2 Create database: `CREATE DATABASE water_supply_india;`
- 3 Load dim tables first (small), then fact tables using Python:
`df.to_sql('dim_cities', engine, if_exists='replace')`
- 4 Add foreign key constraints and composite indexes after loading: `CREATE INDEX idx_city_date ON fact_demand(city_id, date);`
- 5 Connect Power BI → Get Data → MySQL Database → enter localhost credentials



MongoDB Storage Stories

US-03, 05, 06, 09, 12, 16, 19, 24

- 1 Install MongoDB Community + MongoDB Compass (free GUI) from mongodb.com
- 2 Install Python driver: `pip install pymongo`
- 3 Load CSV into MongoDB: `df.to_dict('records')` then
`collection.insert_many(records)`
- 4 Create indexes: `collection.create_index([('city_id',1), ('source_type',1)])`
- 5 Query example for US-03: `collection.find({'is_active': True, 'current_capacity_mld': {'$lt': 0.7}})`



Recommended Implementation Order

Do this sequence to avoid blockers

- 1 **Preprocessing** — run `preprocess_water_supply.py` → generates all cleaned CSVs
- 2 **MySQL setup** — load all dim_ and fact_ tables → foundation for Power BI
- 3 **PySpark EDA + ML** — run `pyspark_pipeline.py` → covers US-01, 06, 09, 10, 11, 12
- 4 **Power BI dashboards** — connect to MySQL → build US-02, 07, 14, 17, 20, 25
- 5 **MongoDB** — store alerts, ML outputs, disruption audit → covers US-03, 05, 06, 16, 24

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